"INDIA Is An UNTAPPED MARKET In Terms Of ELECTRONICS REACH And The Potential Is Enormous"

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How has the pandemic changed the game for semicon majors like Renesas? How do they view the acquisition of Arm by Nvidia? To get answers to these questions, Rahul Chopra spoke to Dr Sailesh Chittipeddi, EVP, GM of Renesas' IoT and Infrastructure Business Unit. Here are the extracts of their discussion

Can you share some of the key strategies that have been outlined after Covid-19 and how things have changed?

One of the biggest trends that we see is that there is a big push towards more bandwidth consumption. For example, Netflix or web consumption, Facebook or even text messages—most of these have gone up by over double-digit percentages in a matter of months. That has driven up quite a bit of data centre spend for major players like Google and Amazon.

There has been an increased focus on healthcare and the environment, whether it is for lowering the SpO_2 level or measuring it. There is a correlation between the SpO_2 level and Covid, if your oxygen levels drop alarmingly. So, that has driven interest especially for bio-sensing and ventilator systems.

With the move towards electric vehicles, less polluting activities have taken hold. On the industrial side, certainly smart metering brings more efficient energy consumption. Surveillance and voice user interface have been big and will grow as people are moving away from touch buttons. So, these are some major trends we have seen. Fundamentally, the push is towards data-centric, all the way from front-end to back-end.

Has that resulted in launch of new products also, as in, some products being pushed and others being put on the bench, or has the usual production increased as per greater demand?

A little bit of both. The demand has certainly increased. The need for laptops and personal computing has really gone up dramatically. Even in the USA, there are a lot of regions where school children cannot afford a laptop or a notebook. So, there has been a big push for Chromebooks, for which districts have been placing big orders.

In terms of investment, we have emphasised a little bit more on things like voice and surveillance and are trying to get some of the products out a little bit quicker. For example, we have pushed our 48V scooter and e-bike solutions a little more aggressively than we would have in the past. Same for the ventilator. So, I would not say that we have slowed down on anything per se, instead we accelerated somewhat to take advantage of some of the six scenarios that we are seeing.

How did Renesas manage to launch products so rapidly during Covid-19, while most brands have been silent? What best practices did you follow to ensure new innovative products kept coming out?

We were trying to make up for the lost time. We were late to launch the Renesas Advanced (RA) Arm-based microcontroller family in 2018. So, we launched that MCU family in October 2019 and followed it up rapidly with a whole set of RA MCU products to significantly expand our portfolio. And then we went out and addressed the gaps that we had in our portfolio early-on in a hurry.

One of my messages to the team has been that customers will always remember how you have served them during challenging times. Because challenging times always allow for the formation of best relationships, whether it's between people, or between organisations. Another message has been that this was the time to out-serve the competitors, whether through new products or a better supply chain.

Coming to Arm being acquired by Nvidia, how is that affecting the Renesas strategy now?

First, I would say there is a finite possibility against the deal occurring as the regulatory authorities may have concerns. But assuming it happens, and based on the discussions we've had with Arm, they have been pretty clear that they are still going to be on an open-licensing model.

We recently announced our first implementation of RISC-V-based solutions, which will begin sampling to customers in the second half of 2021. These MCUs are going to be application-specific solutions through which we will attack motor control and voice markets. Beyond that, we have proprietary 8-, 16-, and 32-bit MCUs as well. We have the RX and RL78 families, which have been around for a long time. So, our customers do not have anything to fear major disruption.

On the other side, there are other GPU providers, such as Imagination Technologies, who do provide cores. So, if we need to pivot, we could on the MPU side. But we don't feel a compelling need to do so at this point of time because Arm as well as Nvidia have been very clear that both will support an open-licensing model. But, obviously, you don't want to be caught working with someone with whom you are competing in another area. That brings a bit of a disadvantage. So, we need to be careful. I must add that we had already taken up some steps to come up with some RISC-V-based products, even before the Nvidia news was announced.

That brings up the question, how do you see the future of RISC-V?

I think people wanted an open-source architecture anyway, primarily because the licensing fees were going